ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE H · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Mechatronics paper 1 · practice h.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Accept other correct answers and wording. Circuits and symbols may be drawn in different but correct ways: mark each required element and correct connection.

QUESTION 1

1.1 A three-terminal semiconductor device (collector, base, emitter) in which a small base current controls a larger collector current ✓✓. Uses ✓✓: as a switch; as an amplifier. (4)

1.2 An analogue signal changes smoothly and can have any value within a range ✓ (e.g. a temperature sensor's voltage) ✓, sketch: a smooth curve / sine wave ✓. A digital signal has only two levels, HIGH and LOW (1 and 0) ✓ (e.g. a switch or a PLC input) ✓, sketch: a square wave ✓. (6)

1.3 NPN transistor symbol ✓; base resistor from the control input to the base ✓; relay coil between + supply and collector ✓; emitter to 0 V ✓; diode across the coil with its cathode to + ✓; labels ✓ (6)

1.4 Any TWO, ✓✓ each: changes are made in software without rewiring; fewer parts, smaller and cheaper; no mechanical contacts to wear out; can do timing, counting and calculations easily; easy fault-finding and monitoring; can communicate with other systems. (4)

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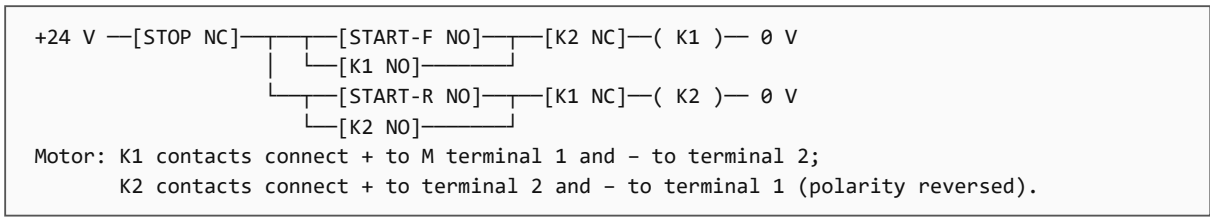
QUESTION 2

2.1 The transmitter changes the operator's commands into a radio signal ✓ and sends it out through its antenna ✓; the receiver's antenna picks up the signal ✓ and the receiver decodes it into control signals for the motors / servos ✓. (4)

2.2 The stator has several coils (phases) ✓; the driver switches the coils on one after another in a set sequence ✓; each pulse moves the magnetic field one position, and the rotor's teeth / magnet line up with it ✓, so the shaft turns a fixed angle (step, e.g. 1,8°) for every pulse ✓. (4)

2.3 Any TWO differences, ✓✓ each: AC motors run from an alternating supply and DC motors from a direct supply; DC motors have a commutator and brushes, most AC (induction) motors do not; DC motor speed is easily controlled by voltage, AC motor speed by frequency (VSD); AC induction motors need less maintenance. (4)

2.4 Common STOP (NC) in the supply ✓; START-F (NO) with K1 holding contact in parallel ✓ in series with NC K2 contact ✓ driving K1 ✓; START-R (NO) with K2 holding contact ✓ in series with NC K1 contact ✓ driving K2 ✓; K1 and K2 main contacts reversing the motor polarity ✓, e.g. (8)



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QUESTION 3

3.1.1 Mass per unit volume of a substance (kg/m³) ✓ (1)

3.1.2 A chemical added to the oil to improve its properties (e.g. anti-wear, anti-foam, anti-rust) ✓ (1)

3.1.3 The forming and collapsing of vapour bubbles in the oil when the pressure (usually at the pump inlet) drops too low, which damages the pump ✓ (1)

3.2.1 Ball (circle) in a V-seat ✓ with the flow lines, free flow in one direction only ✓ (2)

3.2.2 Two squares (2 positions) ✓ with four ports correctly shown (P, T, A, B) ✓; solenoid symbol on one side ✓; spring on the other side ✓ (4)

3.2.3 Diamond (square on its corner) ✓ with a dashed line across it ✓ (2)

3.3 Open centre: in the middle position all ports (or P and T) are connected ✓, so the pump flow returns to tank at low pressure and the cylinder can float / is unloaded ✓. Closed centre: in the middle position all ports are blocked ✓, so the cylinder is held in place and the pump flow goes over the relief valve (or to other branches) ✓. (4)

3.4 The rod-end port is connected to the cap-end port (and to the pump) during extension ✓✓; the oil pushed out of the rod end is added to the pump flow going into the cap end ✓, so the cylinder extends faster ✓, but with less force because the pressure only acts on the area of the rod ✓. (5)

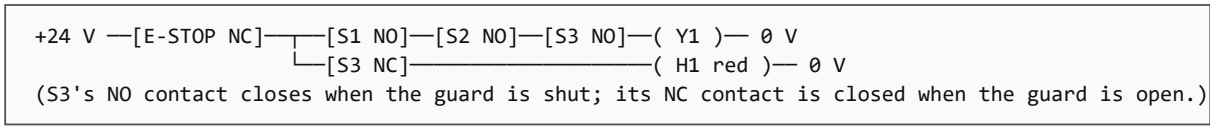
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QUESTION 4

4.1 Information about the actual output (position, speed, pressure) is measured by sensors ✓ and sent back to the controller so it can compare it with the wanted value and correct the system ✓. (2)

4.2 It switches an electrical contact when the hydraulic pressure reaches a set value ✓, e.g. to confirm that a part is clamped before the next step, or to stop the pump at maximum pressure ✓. (2)

4.3 Supply lines ✓; E-stop (NC) in the main supply ✓; S1 (NO) ✓ and S2 (NO) ✓ in SERIES ✓ with guard switch S3 (closes when guard shut) ✓ switching Y1 (or a relay K1 that switches Y1) ✓; no holding contact, so the press retracts when a button is released ✓; lamp H1 switched by an NC contact of S3 (or a relay) so it is on while the guard is open ✓; correct symbols and neat ✓, e.g. (10)



4.4 Any FOUR ✓✓✓✓: the program can be changed without rewiring; far less wiring and smaller panels; timers and counters are built in; easy to find faults (status lights, online monitoring); more reliable (few moving contacts); can communicate with HMIs and computers; cheaper for complex systems. (4)

4.5 It controls the oil flow (and so the position, speed or force of an actuator) smoothly and accurately in proportion to an electrical signal ✓✓, usually with feedback. (2)

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QUESTION 5

5.1 Finding a fault in a logical, step-by-step way ✓, using information, diagrams and tests to narrow down the cause, instead of replacing parts by guessing ✓. (2)

5.2 Connect the gauge at test points in the circuit ✓ and compare the readings with the expected (normal) pressures ✓; low pressure at the pump outlet points to a worn pump, a leak or a relief valve set too low or stuck open ✓; a pressure drop between two points shows a blocked filter or a restriction / internal leakage in the component between them ✓. (4)

5.3 In a logical order ✓✓✓✓: check that the main supply / isolator is on; check fuses and circuit breakers; check that the E-stop is released and the overload is reset; check the control voltage at the START button and the contactor coil; check the contactor and the motor terminals and windings. (4)

5.4 Any FIVE ✓✓✓✓✓: tell the people affected that the machine will be shut down; switch off the machine normally; open the isolator; fit your own lock to the isolator and keep the key; attach a tag with your name, date and reason; release stored energy (pressure, springs, capacitors, raised loads); test that the machine cannot start (try the START button) before working; after the work, remove the tag and lock yourself only. (5)

5.5 Any FIVE ✓✓✓✓✓: check the oil level in the tank; look for oil leaks at hoses, fittings and cylinders; check hoses for wear, cracks or bulges; check that guards are in place; check that the E-stop and safety devices work; check the oil temperature and filter indicator; listen for unusual noise when started; check the pressure gauge reading. (5)

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TOTAL: 100

END OF MARKING GUIDELINE